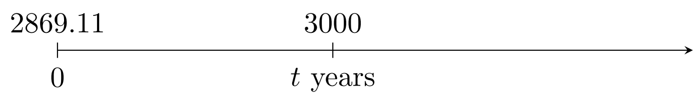
Problem 8.21

A discounted loan of \$3000 at a simple discount rate of 6.5% is offered to Mr. Jones. If the actual amount of money that Mr. Jones receives is \$2869.11, when is the \$3000 due to be paid back?

Solution

The loan can be represented on a timeline as follows:



For a simple discount loan,

$$\text{amount received} = F(1 - dt),$$

where F is the face amount, d is the simple discount rate, and t is the time in years. Here,

$$F = 3000, \qquad d = 6.5\% = 0.065.$$

Thus,

$$2869.11 = 3000(1 - 0.065t).$$

Solving for t,

$$\begin{aligned} 0.065t &= 1 - \frac{2869.11}{3000}, \\ t &= \frac{1 - \frac{2869.11}{3000}}{0.065}, \\ t &\approx 0.67123076 \text{ years}. \end{aligned}$$

Converting to days,

$$0.67123076(365) \approx 244.99895.$$

Therefore, the \$3000 is due in approximately

$$\boxed{245 \text{ days}}.$$